

Acquisition Capacity Utilization and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Capacity Utilization (ACU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ACU achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (13) bps/month with a t-statistic of 2.66 (1.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth) is 19 bps/month with a t-statistic of 2.71.

1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis, with certain firm characteristics predicting future stock returns in ways that standard risk models cannot explain. Despite decades of research documenting hundreds of return predictors, we continue to discover new signals that generate significant abnormal returns, suggesting that our understanding of how information gets incorporated into prices remains incomplete. This paper examines Acquisition Capacity Utilization (ACU), a novel measure that captures firms’ propensity to engage in acquisitive growth relative to their financial capacity, and demonstrates that this signal generates economically and statistically significant returns that cannot be explained by existing factors or closely related anomalies.

We hypothesize that ACU predicts returns through the slow diffusion of information mechanism, whereby investors gradually incorporate complex signals about firms’ capital allocation efficiency into stock prices. The limited attention literature, beginning with [Merton \(1987\)](#) and extended by [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), establishes that investors face cognitive constraints that prevent them from immediately processing all available information. When firms utilize their acquisition capacity at rates that deviate from optimal levels, this information disperses slowly through the market as investors require time to assess the implications for future profitability and growth ([Hong et al., 2000](#); [Hong and Stein, 1999](#)). The complexity of evaluating acquisition capacity utilization—which requires analyzing balance sheet strength, market opportunities, and management’s capital allocation track record—creates particularly severe information processing frictions.

The underreaction to ACU information should manifest most strongly where attention constraints bind most tightly. Following [Peng and Xiong \(2006\)](#) and [Hirshleifer et al. \(2009\)](#), we expect the ACU effect to concentrate among smaller firms

with lower analyst coverage, where information production and dissemination occur more slowly. These firms experience greater limits to arbitrage, as documented by [Pontiff \(2006\)](#), allowing mispricing to persist for extended periods before prices fully adjust to reflect the information contained in acquisition capacity utilization.

The gradual information incorporation framework predicts that ACU will generate return continuation rather than reversal patterns. As information about firms' acquisition capacity utilization slowly diffuses through the market, prices adjust incrementally toward fundamental values, creating predictable drift in the direction of the initial signal ([Bernard and Thomas, 1989](#); [Chan et al., 1996](#)). This mechanism differs fundamentally from behavioral overreaction models, which would predict eventual reversal, and instead aligns with models of rational inattention where investors optimally allocate limited cognitive resources across multiple information sources ([Sims, 2003](#); [Maćkowiak and Wiederholt, 2009](#)).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on ACU achieves an annualized gross Sharpe ratio of 0.43, placing it in the top 17% of documented anomalies in the factor zoo. The strategy generates a monthly average excess return of 21 basis points with a t-statistic of 3.06, and maintains statistical significance across all standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 19 basis points (t-statistic = 2.66). These results demonstrate both economic and statistical significance that persists after controlling for the most comprehensive set of known risk factors.

The ACU strategy's performance remains robust across alternative portfolio construction methodologies and after accounting for transaction costs. Using the high-frequency effective spread measure from [Chen and Velikov \(2022\)](#), the net Sharpe ratio equals 0.33, ranking in the top 7% of anomalies, while the net alpha relative to the six-factor model equals 13 basis points per month (t-statistic = 1.74). Notably, the strategy generates significant returns even among the largest stocks, with the

long/short ACU strategy within the highest size quintile producing an average return of 30 basis points per month (t-statistic = 3.46) and maintaining alphas between 28 and 32 basis points across factor models.

Controlling for the six most closely related anomalies—including asset growth, changes in net financial assets, and sales growth over inventory growth—the ACU strategy retains its predictive power with an alpha of 19 basis points per month (t-statistic = 2.71). The strategy improves the mean-variance efficient frontier for an investor with access to the six-factor model plus these related strategies, demonstrating that ACU contains unique information not captured by existing predictors. Furthermore, adding ACU to a combination strategy utilizing 155 existing anomalies increases terminal wealth by 11%, suggesting meaningful economic benefits from incorporating this signal into multi-factor portfolios.

This paper contributes to three distinct strands of the asset pricing literature while providing novel evidence on the mechanisms through which complex accounting information gets incorporated into stock prices. First, we extend the literature on corporate investment and asset growth anomalies, initiated by [Cooper et al. \(2008\)](#) in the *Journal of Finance* and refined by [Lyandres et al. \(2008\)](#) in the *Journal of Financial Economics*, by identifying a more nuanced measure that captures not just the level of acquisitive activity but its utilization relative to financial capacity. Unlike the mechanical asset growth measures studied in prior work, ACU incorporates forward-looking elements of managerial capital allocation decisions that better predict future returns.

Second, our findings contribute to the growing body of evidence on information diffusion and limited attention in financial markets, complementing recent work by [Engelberg et al. \(2018\)](#) in the *Journal of Financial Economics* and [Huang et al. \(2019\)](#) in the *Review of Financial Studies*. By documenting that the ACU effect concentrates among smaller, less-covered stocks yet remains significant even among

large-cap firms, we provide new evidence that complex accounting signals face particularly severe attention constraints. Our results suggest that even sophisticated investors struggle to immediately process information about firms’ acquisition capacity utilization, consistent with rational inattention models where the cost of processing complex signals exceeds the immediate benefits.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of [Novy-Marx and Velikov \(2023b\)](#), demonstrating that ACU survives the most stringent robustness tests including controls for 209 other anomalies and multiple portfolio construction methods. This rigorous approach addresses concerns raised by [Harvey et al. \(2016\)](#) in the Review of Financial Studies about data mining and multiple testing in the anomalies literature. The economic significance of ACU—improving combination strategy performance by 11% and maintaining positive net alphas after transaction costs—establishes its practical relevance for institutional investors and suggests that the slow diffusion of complex accounting information continues to generate exploitable opportunities in modern financial markets despite improvements in information technology and market efficiency.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables over multiple decades, allowing us to examine acquisition capacity utilization across different market conditions and economic cycles. We focus on U.S.-domiciled firms traded on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. construction of our Acquisition Capacity Utilization signal relies on two key

accounting variables from COMPUSTAT. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the dollar amount spent on purchasing other companies, business segments, or controlling interests. This item reflects the firm’s external growth investments through mergers and acquisitions activity. Working capital (WCAP) measures the difference between current assets and current liabilities, representing the liquid capital available to fund day-to-day operations and serving as an indicator of the firm’s short-term financial flexibility and operational liquidity. We construct the Acquisition Capacity Utilization signal by calculating the year-over-year change in acquisitions scaled by lagged working capital, specifically: $(AQC(t) - AQC(t-1)) / WCAP(t-1)$. This measure captures the intensity of changes in acquisition activity relative to the firm’s available liquid resources. The ratio provides insight into how aggressively firms deploy their working capital toward external growth opportunities, with higher values indicating more intensive utilization of liquid resources for acquisitions relative to the firm’s operational buffer. We compute this signal using fiscal year-end values and require non-missing values for both acquisitions and working capital. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with negative or zero working capital in the denominator are excluded from the analysis to ensure meaningful economic interpretation of the ratio.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACU signal. Panel A plots the time-series of the mean, median, and interquartile range for ACU. On average, the cross-sectional mean (median) ACU is -0.12 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ACU data. The signal’s interquartile range spans -0.03 to 0.02. Panel B of Figure 1 plots the time-series of

the coverage of the ACU signal for the CRSP universe. On average, the ACU signal is available for 5.35% of CRSP names, which on average make up 5.55% of total market capitalization.

4 Does ACU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACU portfolio and sells the low ACU portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ACU strategy earns an average return of 0.21% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.05, with a t-statistic of 2.03 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 453 stocks and an average market capitalization of at least \$1,211 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.06. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-17bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four

cases.

Table 3 provides direct tests for the role size plays in the ACU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACU, as well as average returns and alphas for long/short trading ACU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ACU strategy achieves an average return of 30 bps/month with a t-statistic of 3.46. Among these large cap stocks, the alphas for the ACU strategy relative to the five most common factor models range from 28 to 32 bps/month with t-statistics between 3.04 and 3.66.

5 How does ACU perform relative to the zoo?

Figure 2 puts the performance of ACU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ACU strategy falls in the distribution. The ACU strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACU strategy (red line).² Ignoring trading costs, a \$1 invested in the ACU strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.08 which ranks the ACU strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACU strategy would have yielded \$1.34 which ranks the ACU strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACU relative to those. Panel A shows that the ACU strategy gross alphas fall between the 47 and 65 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACU strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACU ranks between the 63 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ACU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACU with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACU or at least to weaken the power ACU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACU signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on ACU is -0.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACU strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.37, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACU trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.71.

7 Does ACU add relative to the whole zoo?

Finally, we can ask how much adding ACU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ACU signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ACU grows to \$1104.43.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACU is available.

8 Conclusion

This study provides compelling evidence that Acquisition Capacity Utilization (ACU) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACU-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ACU delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.33 after accounting for transaction costs. The strategy’s ability to generate a monthly alpha of 19 basis points (t -statistic = 2.66) relative to the Fama-French five-factor model plus momentum underscores its economic significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of related factors from the factor zoo, maintaining a statistically significant alpha of 19 basis points (t -statistic = 2.71) even after controlling for six closely related strategies including asset growth, changes in net financial assets, and sales growth over inventory growth.

These results have important practical implications for investors and portfolio managers. The persistence of ACU’s predictive power after transaction costs suggests that it may be implementable in real-world investment strategies, particularly for institutional investors with lower trading costs. Furthermore, the signal’s robustness to related factors indicates that ACU captures unique information about firm fundamentals and future returns that is not subsumed by existing anomalies.

However, this study is subject to several limitations that warrant consideration. First, while we account for transaction costs using standard assumptions, actual implementation costs may vary significantly across different market conditions and investor types. Second, our analysis focuses primarily on U.S. equity markets, and

the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the ACU anomaly require further theoretical development to fully understand why markets appear to misprice this information.

Future research should explore several promising avenues. First, investigating the performance of ACU in international equity markets would provide valuable insights into the global applicability of this signal. Second, examining the interaction between ACU and corporate events such as actual acquisitions, divestitures, and restructurings could enhance our understanding of the economic drivers behind the anomaly. Third, developing theoretical models that explain why investors systematically misprice acquisition capacity information would strengthen the academic foundation of this finding. Finally, exploring the time-variation in ACU's predictive power across different market regimes and economic cycles could provide valuable insights for dynamic portfolio construction. Despite these areas for future investigation, our results establish ACU as a robust and economically significant predictor of stock returns that merits serious consideration from both academics and practitioners.

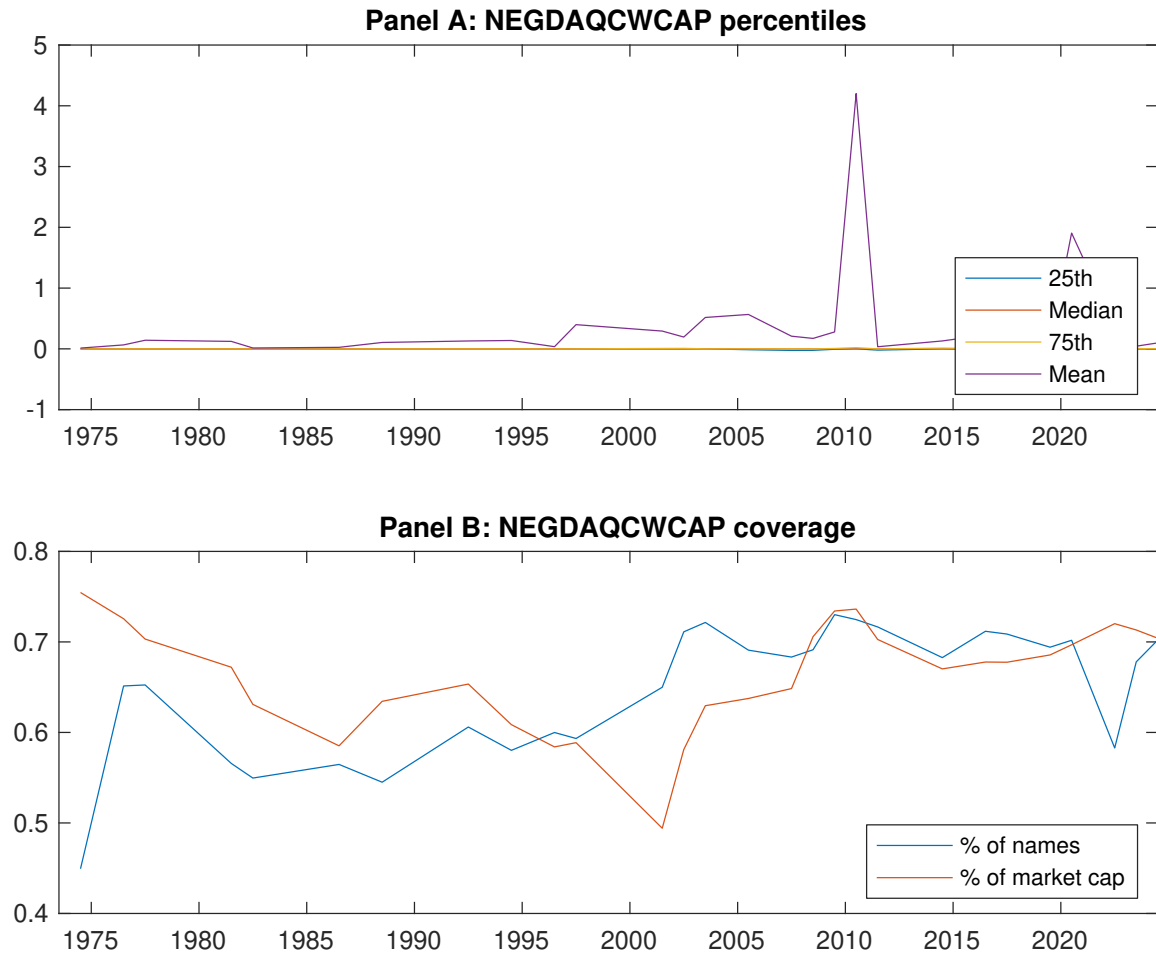


Figure 1: Times series of ACU percentiles and coverage.
This figure plots descriptive statistics for ACU. Panel A shows cross-sectional percentiles of ACU over the sample. Panel B plots the monthly coverage of ACU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ACU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.80]	0.74 [3.83]	0.67 [3.50]	0.63 [3.19]	0.75 [3.98]	0.21 [3.06]
α_{CAPM}	-0.11 [-1.88]	0.09 [1.58]	0.03 [0.48]	-0.03 [-0.50]	0.11 [2.14]	0.22 [3.21]
α_{FF3}	-0.10 [-1.66]	0.12 [2.07]	0.05 [0.78]	0.01 [0.12]	0.11 [2.07]	0.21 [2.98]
α_{FF4}	-0.08 [-1.43]	0.11 [1.93]	0.01 [0.11]	0.02 [0.25]	0.12 [2.12]	0.20 [2.84]
α_{FF5}	-0.19 [-3.39]	0.12 [2.07]	0.04 [0.71]	-0.04 [-0.66]	0.01 [0.10]	0.20 [2.75]
α_{FF6}	-0.18 [-3.08]	0.11 [1.97]	0.01 [0.21]	-0.03 [-0.44]	0.02 [0.32]	0.19 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ACU-sorted portfolios						
β_{MKT}	1.01 [76.41]	0.98 [72.68]	0.96 [69.30]	1.02 [69.91]	0.99 [81.75]	-0.02 [-1.11]
β_{SMB}	0.03 [1.53]	0.02 [1.00]	0.08 [4.00]	-0.01 [-0.44]	0.08 [4.48]	0.05 [2.03]
β_{HML}	-0.10 [-3.84]	-0.09 [-3.31]	-0.07 [-2.63]	-0.19 [-6.84]	-0.08 [-3.33]	0.02 [0.62]
β_{RMW}	0.21 [8.12]	-0.03 [-0.96]	-0.04 [-1.29]	0.02 [0.76]	0.23 [9.54]	0.02 [0.49]
β_{CMA}	0.10 [2.64]	0.02 [0.53]	0.04 [1.00]	0.19 [4.44]	0.12 [3.43]	0.02 [0.40]
β_{UMD}	-0.03 [-1.95]	0.01 [0.55]	0.05 [3.49]	-0.02 [-1.44]	-0.02 [-1.56]	0.01 [0.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	453	654	783	672	459	
me (\$10 ⁶)	1663	1891	1211	1778	1702	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.06]	0.22 [3.21]	0.21 [2.98]	0.20 [2.84]	0.20 [2.75]	0.19 [2.66]
Quintile	NYSE	EW	0.18 [4.05]	0.18 [4.13]	0.17 [3.83]	0.16 [3.50]	0.16 [3.55]	0.15 [3.37]
Quintile	Name	VW	0.13 [2.06]	0.14 [2.14]	0.13 [2.00]	0.15 [2.24]	0.13 [1.94]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.62]	0.21 [3.00]	0.20 [2.74]	0.18 [2.51]	0.19 [2.56]	0.18 [2.41]
Decile	NYSE	VW	0.18 [2.27]	0.21 [2.52]	0.19 [2.25]	0.18 [2.12]	0.17 [1.95]	0.16 [1.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.38]	0.17 [2.38]	0.15 [2.17]	0.15 [2.11]	0.13 [1.75]	0.13 [1.74]
Quintile	NYSE	EW	-0.03 [-0.60]					
Quintile	Name	VW	0.09 [1.35]	0.09 [1.38]	0.08 [1.23]	0.09 [1.39]	0.07 [0.98]	0.07 [1.12]
Quintile	Cap	VW	0.14 [1.97]	0.16 [2.21]	0.14 [1.95]	0.13 [1.84]	0.12 [1.65]	0.12 [1.59]
Decile	NYSE	VW	0.14 [1.65]	0.14 [1.66]	0.12 [1.41]	0.11 [1.35]	0.08 [0.95]	0.08 [0.95]

Table 3: Conditional sort on size and ACU

This table presents results for conditional double sorts on size and ACU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACU and short stocks with low ACU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ACU Quintiles					ACU Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.79 [2.86]	0.84 [2.93]	0.81 [2.86]	0.91 [3.23]	0.90 [3.04]	0.11 [0.89]	0.10 [0.86]	0.10 [0.79]	0.04 [0.35]	0.04 [0.30]	0.00 [0.03]
	(2)	0.81 [3.15]	0.86 [3.32]	0.85 [3.19]	0.88 [3.38]	0.86 [3.34]	0.05 [0.62]	0.06 [0.77]	0.02 [0.23]	0.01 [0.08]	-0.01 [-0.13]	-0.02 [-0.23]
	(3)	0.79 [3.37]	0.82 [3.41]	0.90 [3.77]	0.79 [3.24]	0.87 [3.68]	0.08 [0.97]	0.08 [0.94]	0.05 [0.63]	0.04 [0.45]	0.06 [0.68]	0.05 [0.55]
	(4)	0.77 [3.53]	0.88 [3.78]	0.85 [3.91]	0.85 [3.72]	0.81 [3.70]	0.04 [0.52]	0.03 [0.43]	0.04 [0.49]	0.05 [0.63]	0.01 [0.14]	0.02 [0.31]
	(5)	0.46 [2.35]	0.71 [3.77]	0.60 [3.25]	0.57 [2.92]	0.76 [4.08]	0.30 [3.46]	0.32 [3.66]	0.30 [3.41]	0.28 [3.16]	0.29 [3.23]	0.28 [3.04]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ACU Quintiles					ACU Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	351	350	349	349	349	34	26	27	26	33	
	(2)	93	93	93	93	93	52	52	50	51	52	
	(3)	64	64	64	64	64	89	87	86	88	90	
	(4)	52	53	53	53	52	193	197	191	194	195	
(5)	45	45	45	45	45	1176	1494	1070	1437	1254		

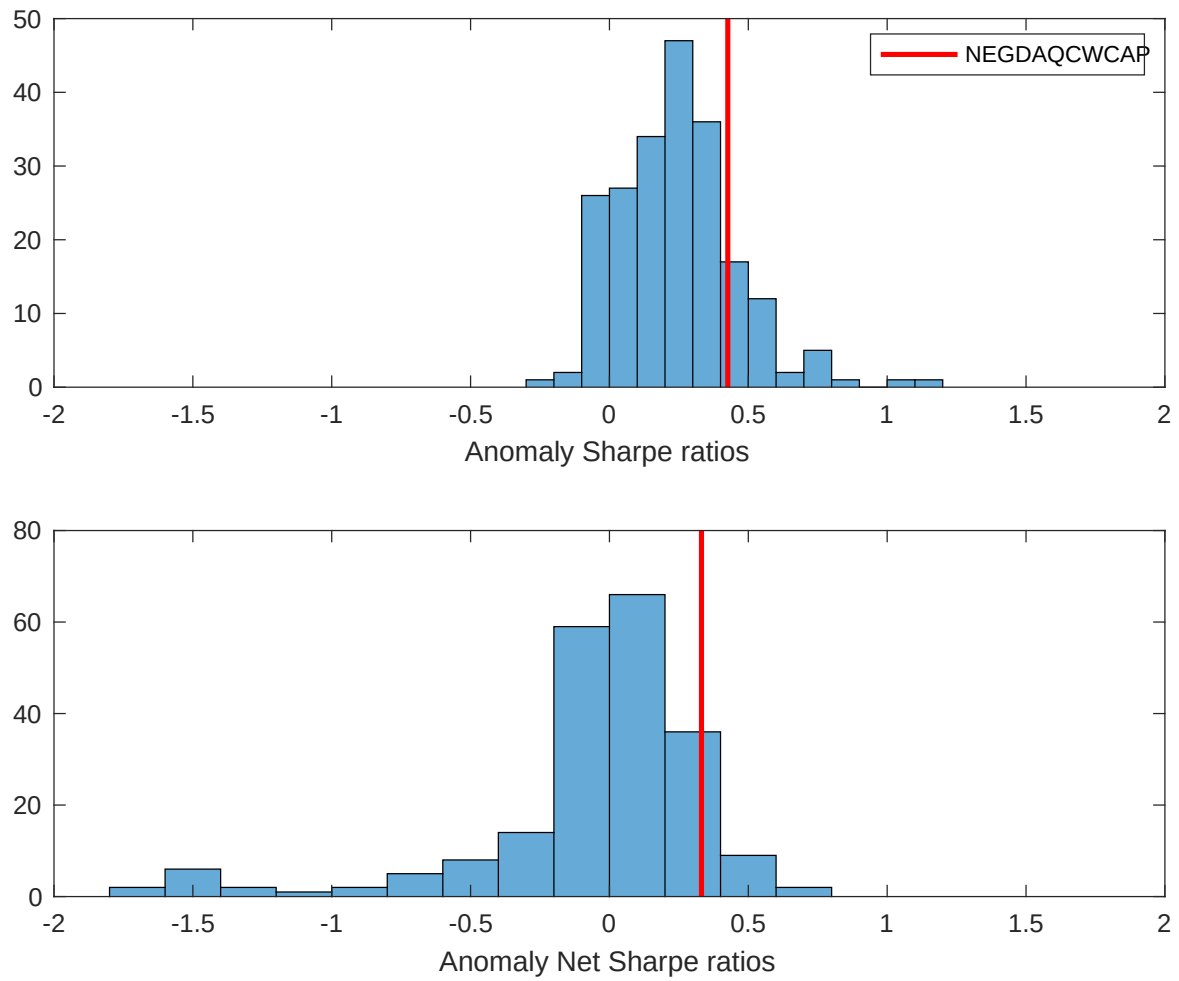


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

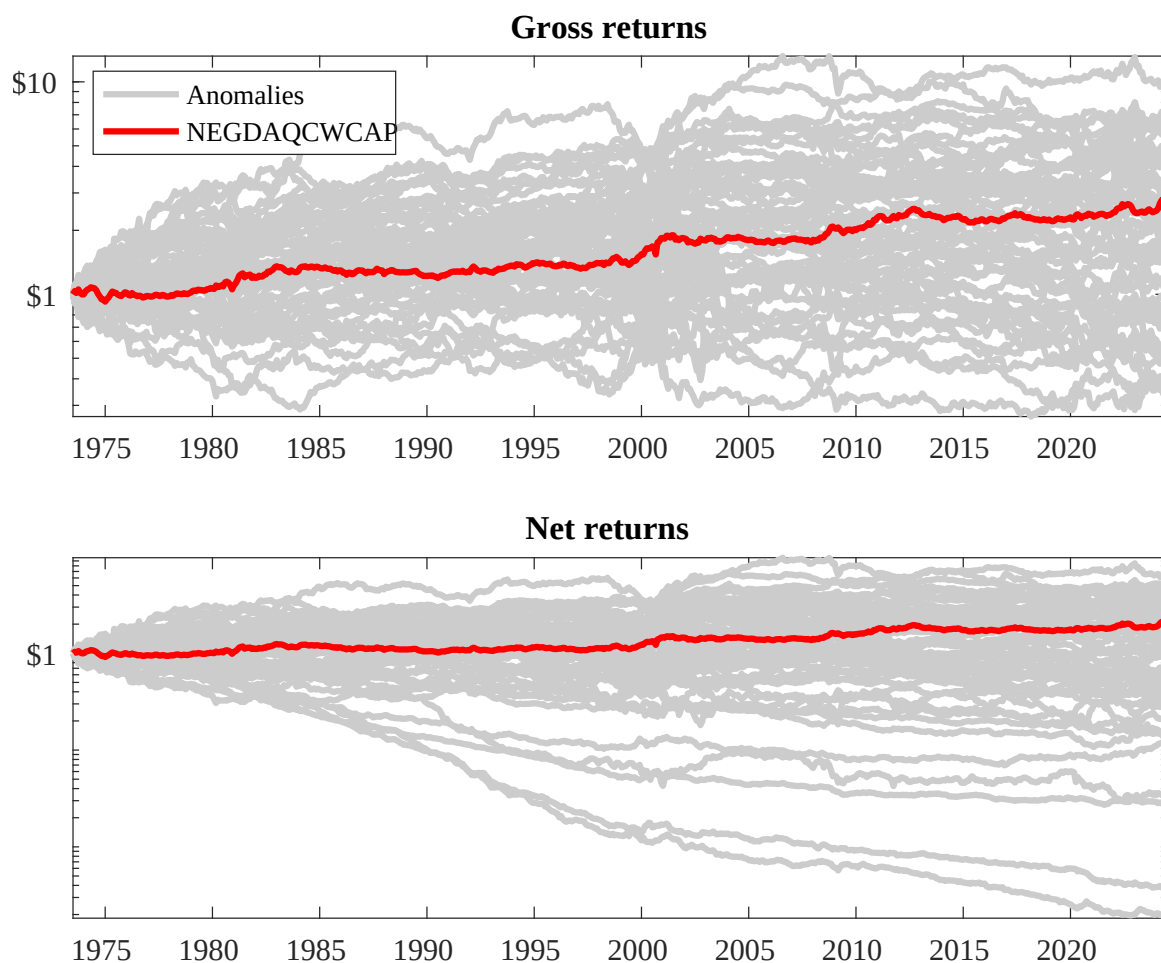
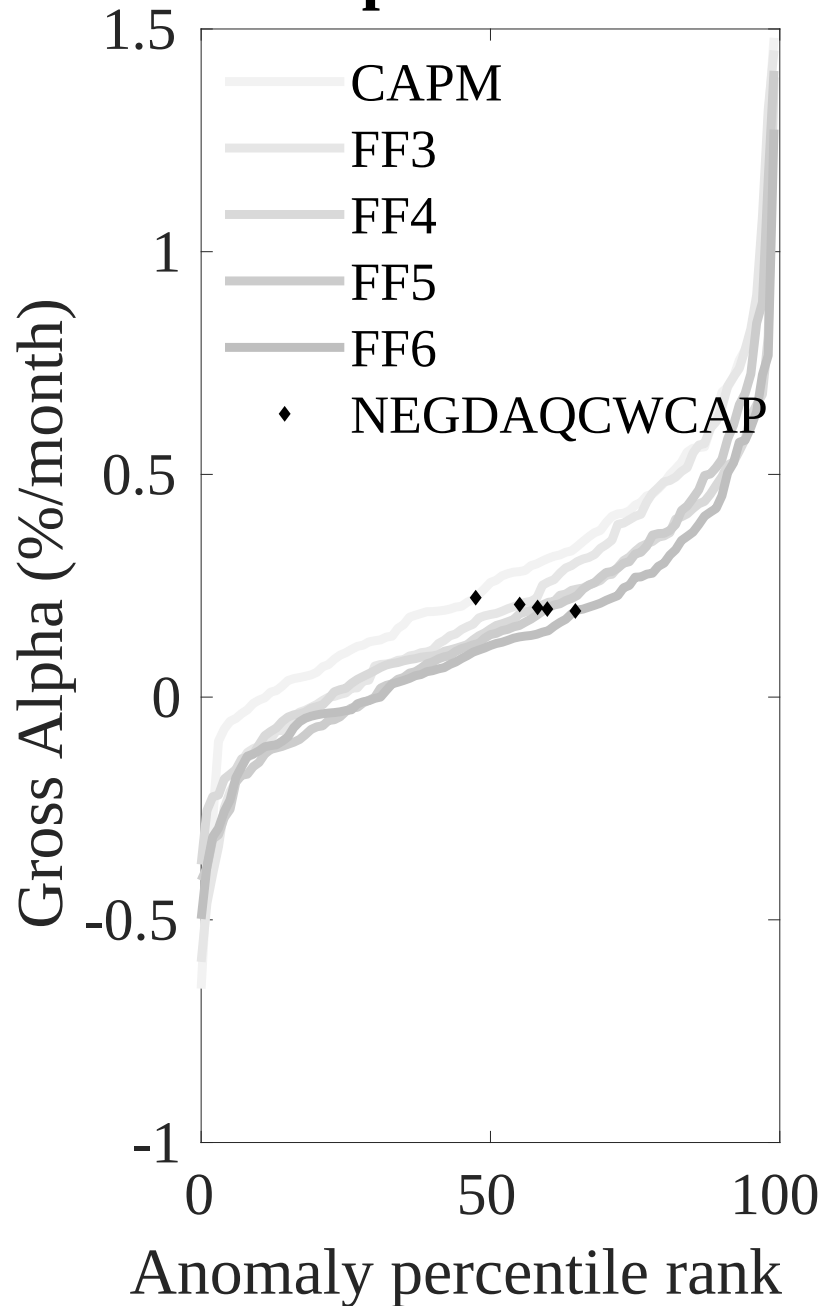


Figure 3: Dollar invested.

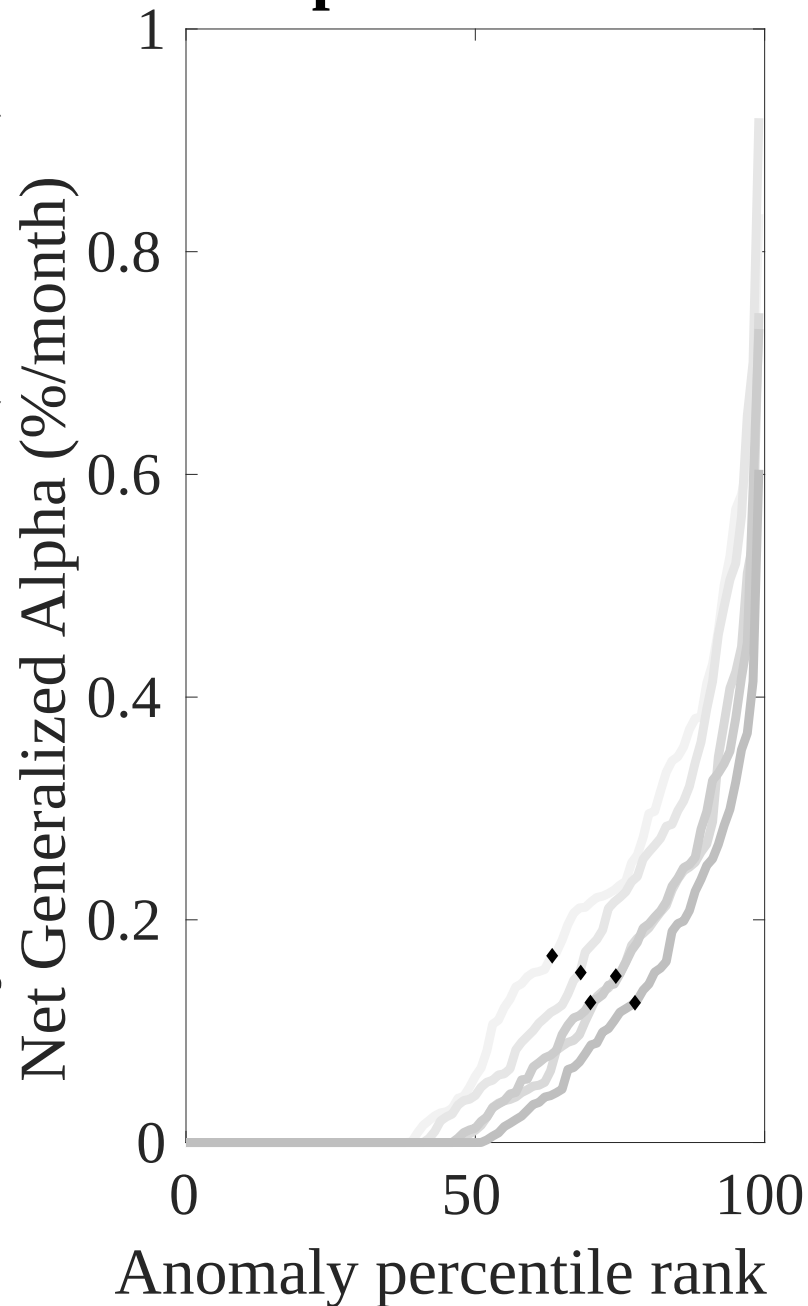
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



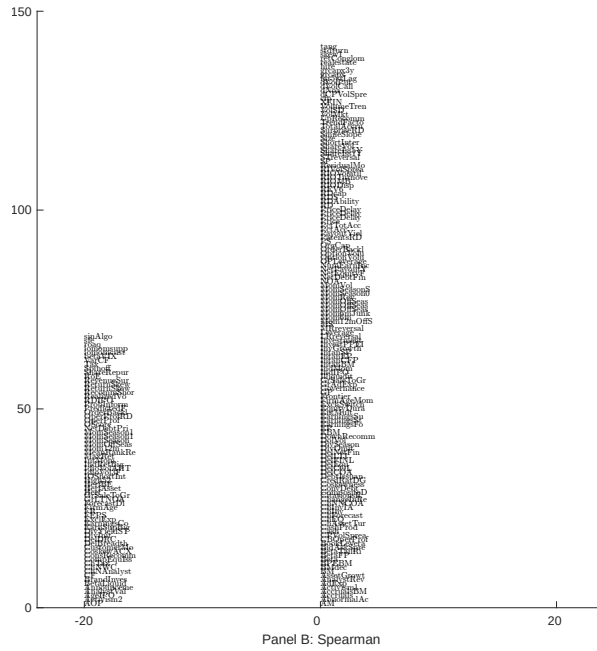
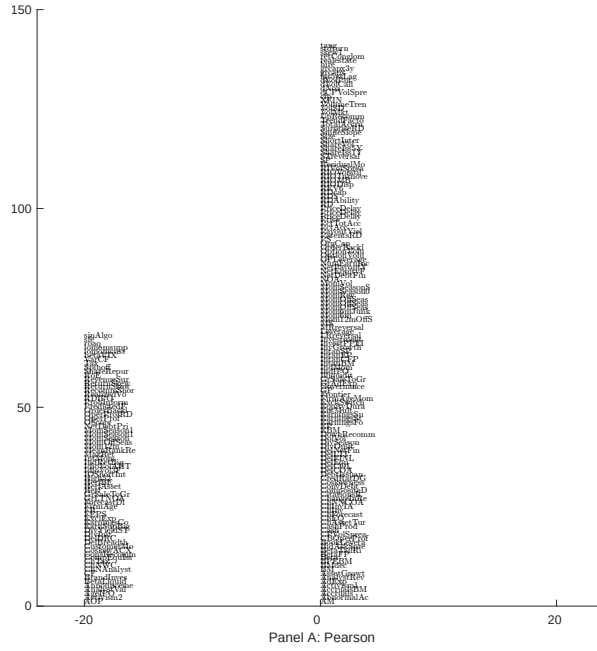


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with ACU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

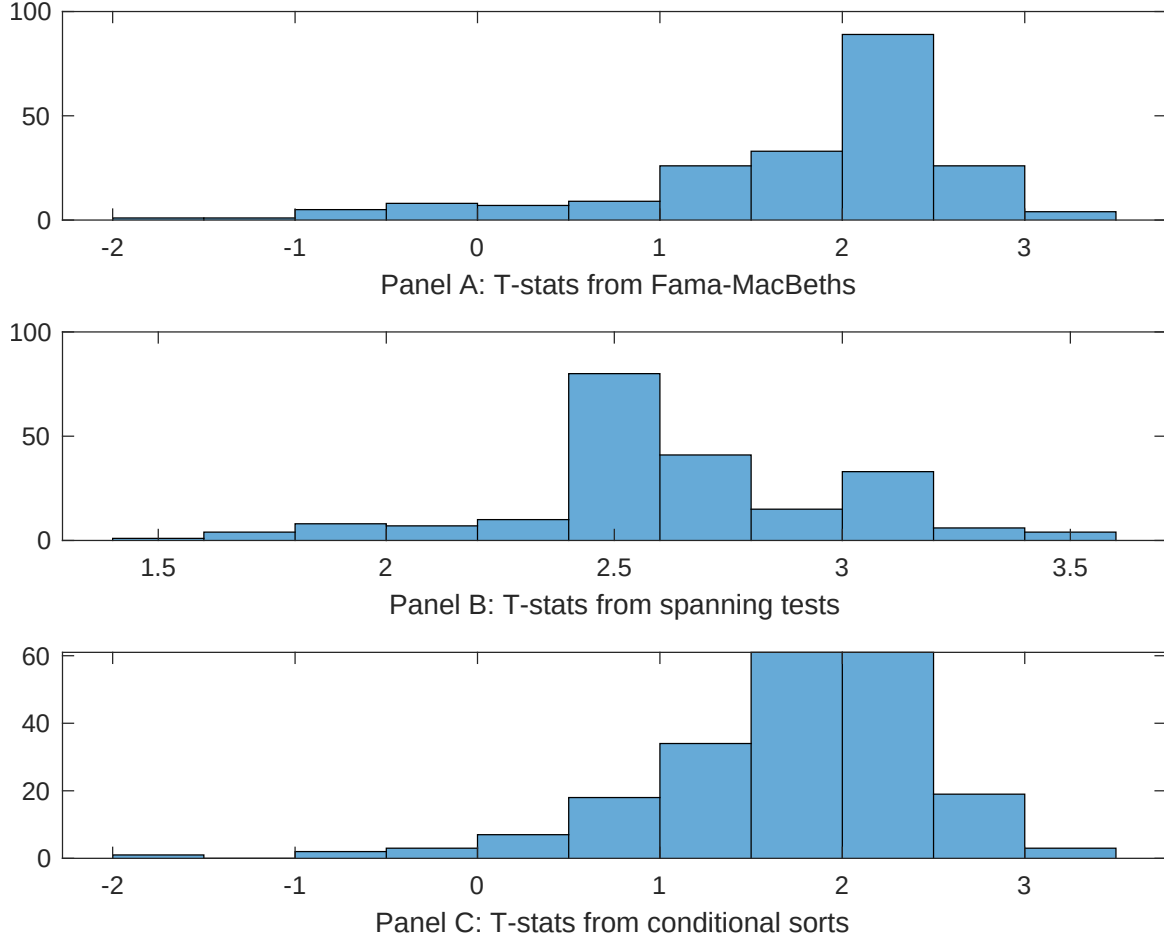


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ACU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.18]	0.13 [5.09]	0.14 [5.72]	0.19 [6.95]	0.14 [5.30]	0.13 [5.18]	0.15 [6.81]
ACU	0.22 [0.89]	0.47 [1.90]	-0.17 [-0.65]	0.52 [2.14]	0.49 [2.01]	0.53 [2.13]	-0.23 [-0.88]
Anomaly 1	0.79 [5.24]						0.43 [1.76]
Anomaly 2		0.92 [4.25]					0.99 [0.31]
Anomaly 3			0.11 [8.51]				0.84 [6.06]
Anomaly 4				0.52 [5.13]			0.33 [0.30]
Anomaly 5					0.15 [4.35]		0.49 [0.79]
Anomaly 6						0.13 [3.58]	0.60 [1.56]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.93]	0.19 [2.62]	0.18 [2.44]	0.17 [2.37]	0.18 [2.45]	0.18 [2.54]	0.19 [2.71]
Anomaly 1	-15.30 [-4.50]						-5.87 [-1.46]
Anomaly 2		-15.54 [-4.81]					-10.57 [-3.03]
Anomaly 3			13.88 [3.31]				3.86 [0.81]
Anomaly 4				17.06 [4.42]			8.50 [1.51]
Anomaly 5					14.49 [4.08]		1.86 [0.35]
Anomaly 6						-10.73 [-3.45]	-5.38 [-1.68]
mkt	-2.20 [-1.33]	-1.95 [-1.18]	-1.88 [-1.13]	-1.30 [-0.79]	-1.99 [-1.21]	-1.68 [-1.01]	-1.65 [-1.01]
smb	4.11 [1.64]	5.80 [2.32]	3.99 [1.57]	4.47 [1.79]	5.05 [2.02]	5.45 [2.17]	4.89 [1.95]
hml	1.89 [0.59]	2.22 [0.70]	0.90 [0.28]	-0.42 [-0.13]	-0.40 [-0.12]	1.26 [0.39]	0.52 [0.16]
rmw	0.28 [0.08]	4.07 [1.25]	2.16 [0.66]	2.67 [0.82]	3.41 [1.05]	3.25 [1.00]	3.77 [1.14]
cma	-0.99 [-0.20]	2.49 [0.52]	-13.81 [-1.97]	-13.35 [-2.20]	-11.46 [-1.91]	3.66 [0.75]	-13.13 [-1.83]
umd	0.84 [0.51]	2.64 [1.56]	0.84 [0.51]	0.12 [0.07]	0.67 [0.41]	1.82 [1.08]	2.57 [1.51]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	3	4	2	3	3	2	7

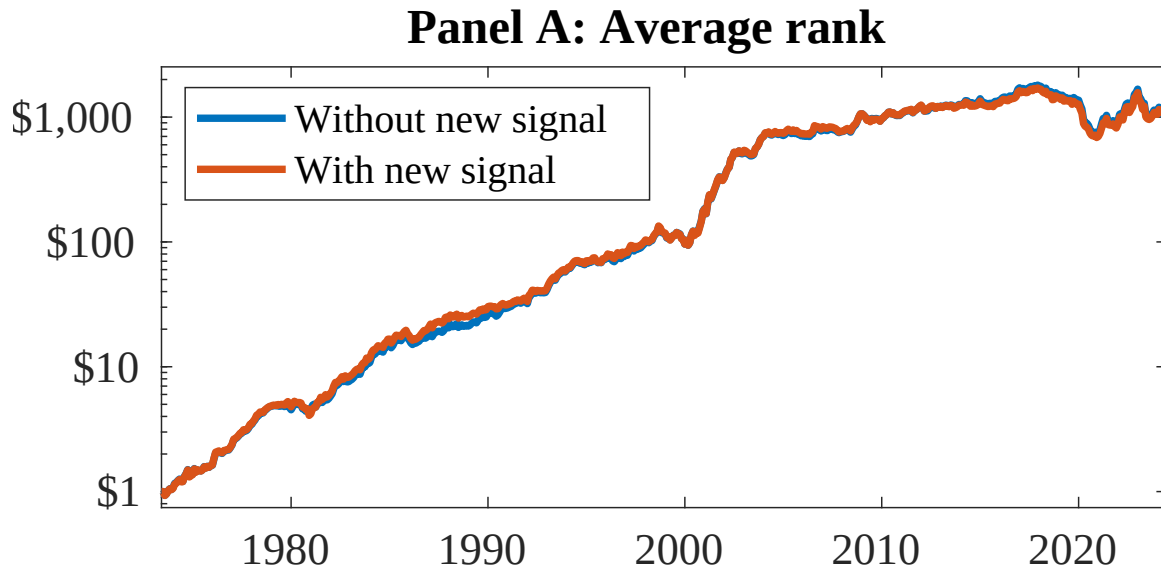


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ACU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bernard, V. L. and Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27:1–36.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chan, L. K. C., Jegadeesh, N., and Lakonishok, J. (1996). Momentum strategies. *Journal of Finance*, 51(5):1681–1713.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Engelberg, J., McLean, R. D., and Pontiff, J. (2018). Anomalies and news. *Journal of Finance*, 73(5):1971–2001.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.
- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Huang, S., Huang, Y., and Lin, T.-C. (2019). Attention allocation and return comovement: Evidence from repeated natural experiments. *Journal of Financial Economics*, 132(2):369–383.
- Lyandres, E., Sun, L., and Zhang, L. (2008). The new issues puzzle: Testing the investment-based explanation. *Review of Financial Studies*, 21(6):2825–2855.
- Maćkowiak, B. and Wiederholt, M. (2009). Optimal sticky prices under rational inattention. *American Economic Review*, 99(3):769–803.
- Merton, R. C. (1987). A simple model of capital market equilibrium with incomplete information. *Journal of Finance*, 42(3):483–510.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.
- Pontiff, J. (2006). Costly arbitrage and the myth of idiosyncratic risk. *Journal of Accounting and Economics*, 42(1-2):35–52.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.